FINITE PRESENTATION OVER A FIRST ORDER THICKENING

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma defos-lemma-deform-fp-module-ringed-topoi (canonical tag 08VU). cited at proof line 1426. Fp modules over a first order thickening.

Source reference. Original lemma 08VU: sources/stacks-chow/tags.txt, source lines 11247–11247. Cited at statement line 1426: (2)<->(4) is a special case.

Lemma 0.2. *Let*

$$\begin{array}{ccc} X & \xrightarrow[i]{} & X' \\ \downarrow & & \downarrow \\ T & \longrightarrow & T' \end{array}$$

be a cartesian square of algebraic spaces where $T \to T'$ is a first order thickening. Let $\mathcal{F}'$ be an $\mathcal{O}_{X'}$-module flat over T'. Set $\mathcal{F} = i^*\mathcal{F}'$. The following are equivalent

(1) *$\mathcal{F}'$ is a quasi-coherent $\mathcal{O}_{X'}$-module of finite presentation,*
(2) *$\mathcal{F}'$ is an $\mathcal{O}_{X'}$-module of finite presentation,*
(3) *$\mathcal{F}$ is a quasi-coherent $\mathcal{O}_X$-module of finite presentation,*
(4) *$\mathcal{F}$ is an $\mathcal{O}_X$-module of finite presentation,*

Proof. Recall that a finitely presented module is quasi-coherent hence the equivalence of (1) and (2) and (3) and (4). The equivalence of (2) and (4) is a special case of Deformation Theory, Lemma 08VU. □

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